

CSE 4125: Distributed Database Systems

Chapter – 6

Optimization of Access Strategies.
(part – C)

Outline

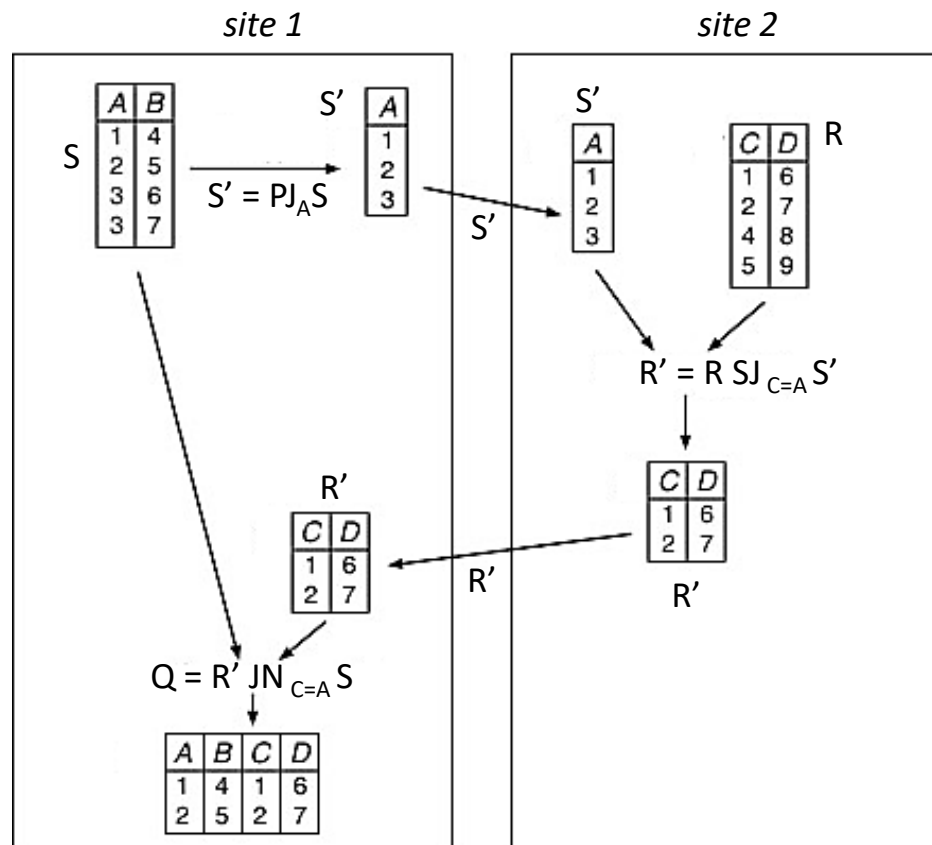
- Semi-join programs.
- Full reducer.

Semi-join Programs

$$R \Join_{C=A} S \leftrightarrow (R \Join_{C=A} \Pi_A S) \Join_{C=A} S$$

Semi-join Programs (contd.)

$$R \text{ JN}_{C=A} S \leftrightarrow (R \text{ SJ}_{C=A} \text{ PJ}_A S) \text{ JN}_{C=A} S$$

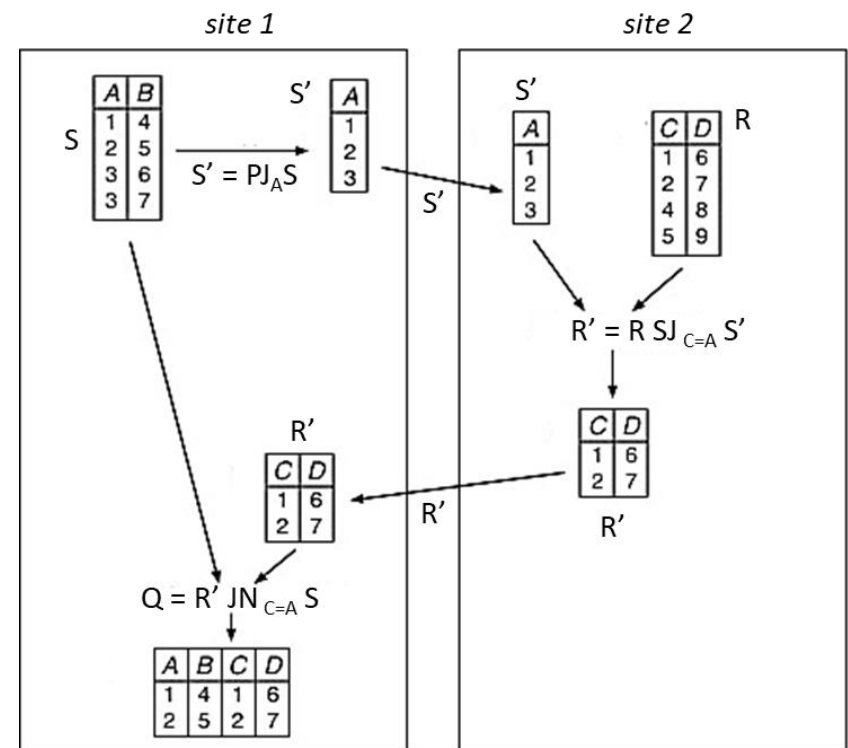


Semi-join Programs (contd.)

Cost of Semi-join program

- Cost of sending S' to site – 2:

$$TC_1 = ?$$

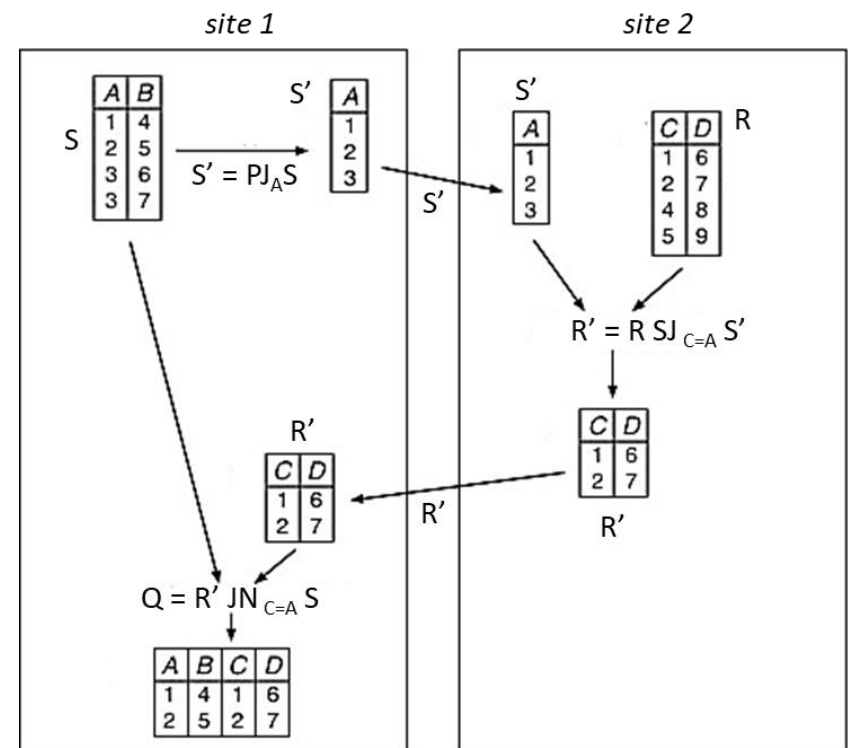


Semi-join Programs (contd.)

Cost of Semi-join program

- Cost of sending S' to site – 2:

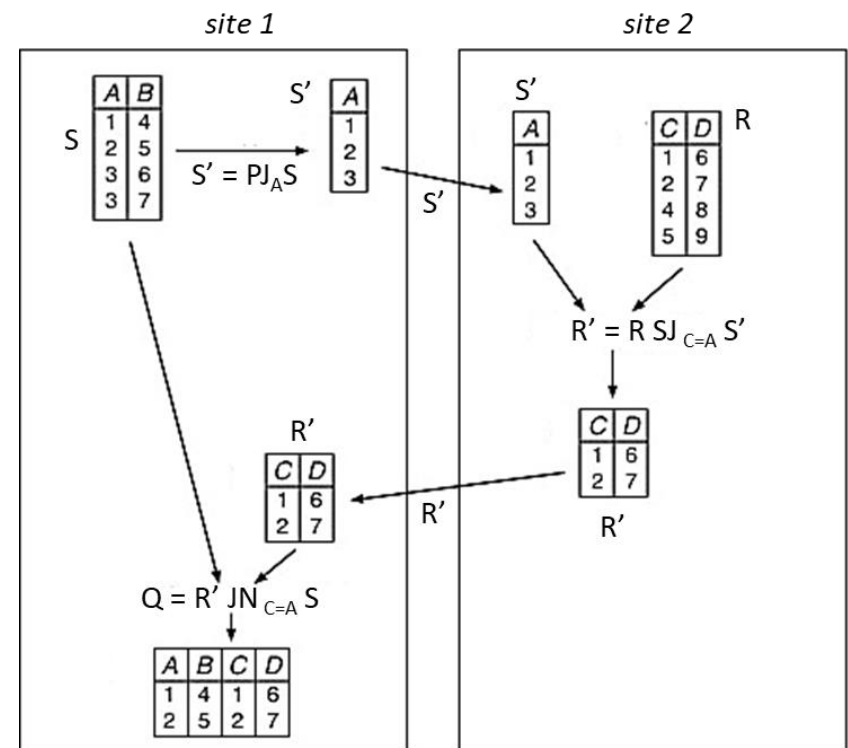
$$TC_1 = C_0 + C_1 * \text{size}(A) * \text{val}(A[S])$$



Semi-join Programs (contd.)

Cost of Semi-join program

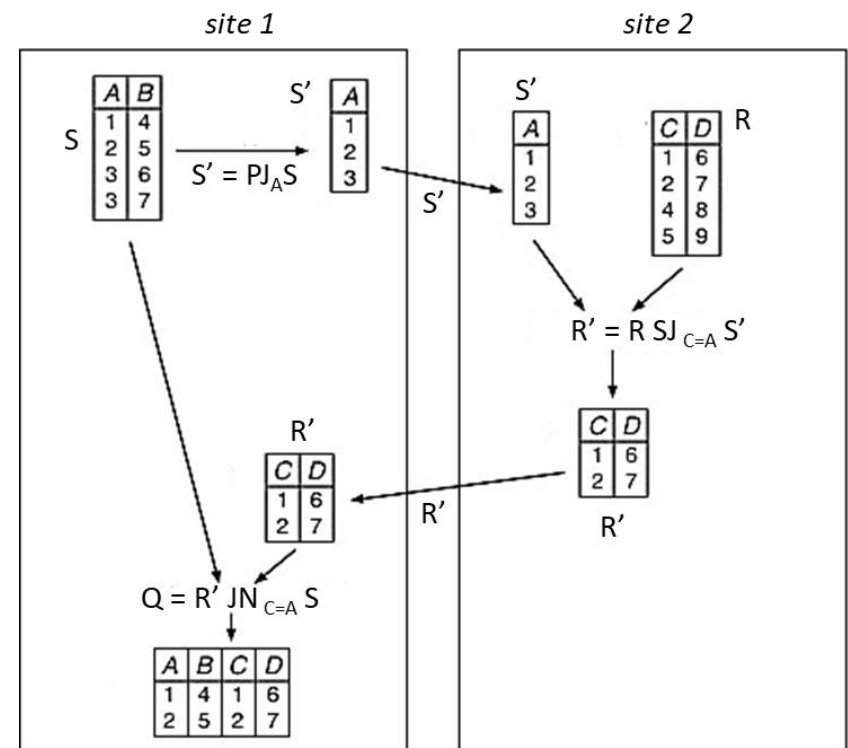
- Cost of sending S' to site – 2:
 $TC_1 = C_0 + C_1 * \text{size}(A) * \text{val}(A[S])$
- Cost of computing R' at site – 2:
 $TC_2 = ?$



Semi-join Programs (contd.)

Cost of Semi-join program

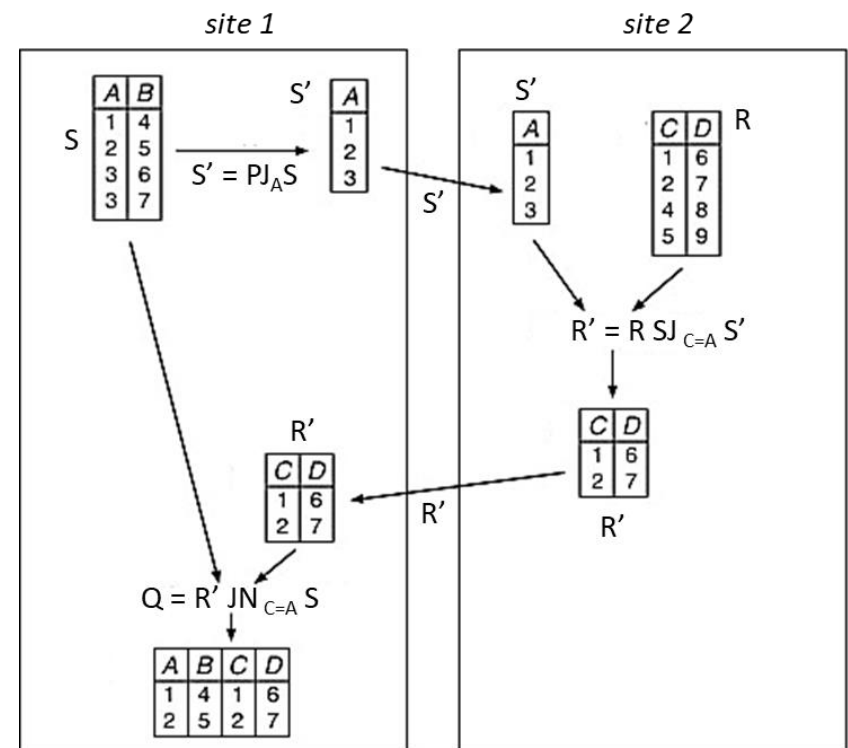
- Cost of sending S' to site – 2:
 $TC_1 = C_0 + C_1 * \text{size}(A) * \text{val}(A[S])$
- Cost of computing R' at site – 2:
 $TC_2 = 0$



Semi-join Programs (contd.)

Cost of Semi-join program

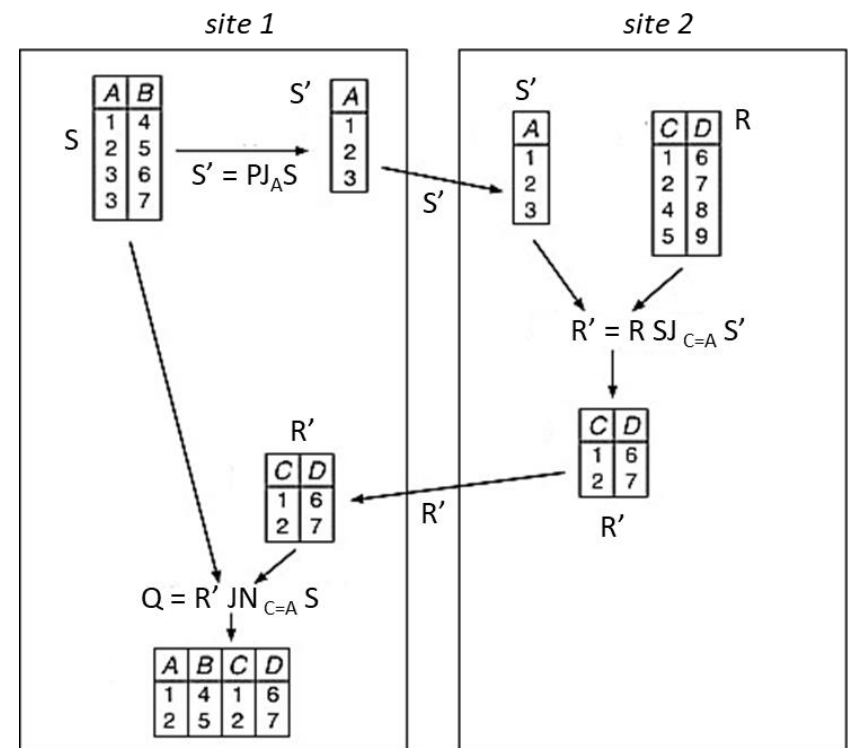
- Cost of sending S' to site – 2:
 $TC_1 = C_0 + C_1 * \text{size}(A) * \text{val}(A[S])$
- Cost of computing R' at site – 2:
 $TC_2 = 0$
- Cost of sending R' to site – 1:
 $TC_3 = ?$



Semi-join Programs (contd.)

Cost of Semi-join program

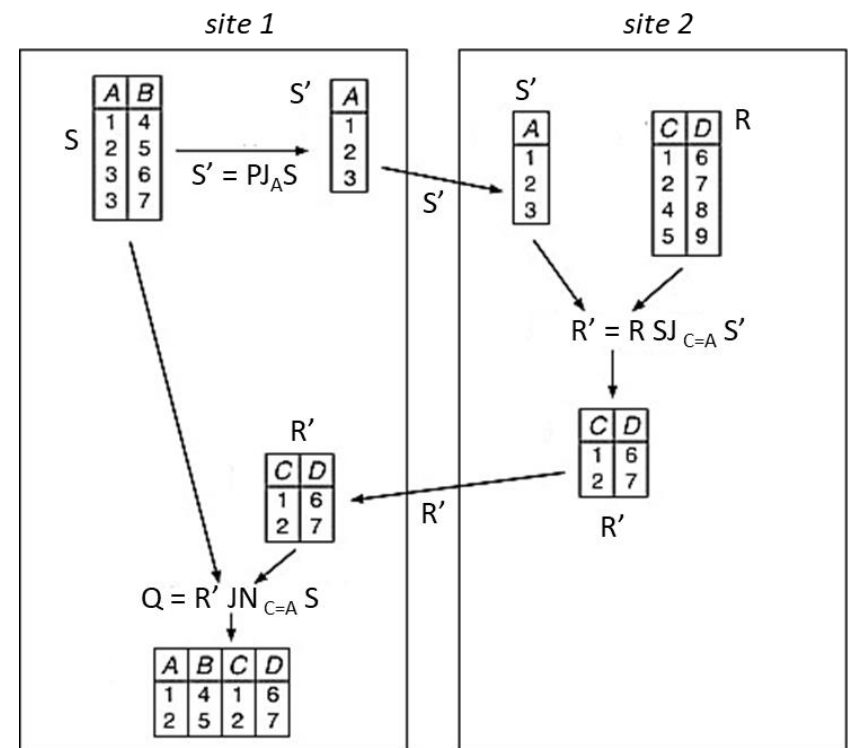
- Cost of sending S' to site – 2:
 $TC_1 = C_0 + C_1 * \text{size}(A) * \text{val}(A[S])$
- Cost of computing R' at site – 2:
 $TC_2 = 0$
- Cost of sending R' to site – 1:
 $TC_3 = C_0 + C_1 * \text{size}(R) * \text{card}(R')$



Semi-join Programs (contd.)

Cost of Semi-join program

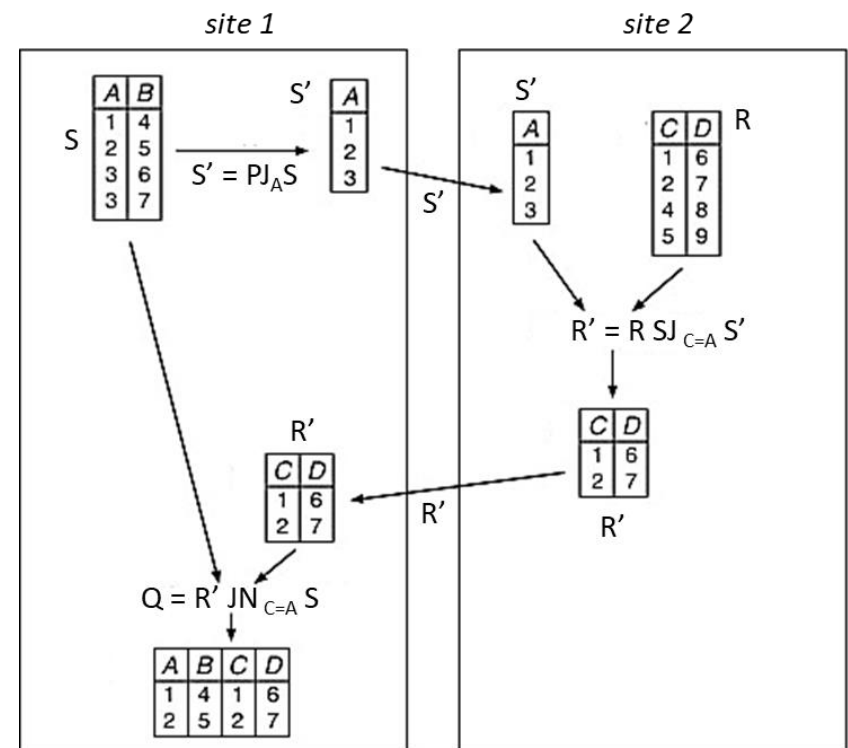
- Cost of sending S' to site – 2:
 $TC_1 = C_0 + C_1 * \text{size}(A) * \text{val}(A[S])$
- Cost of computing R' at site – 2:
 $TC_2 = 0$
- Cost of sending R' to site – 1:
 $TC_3 = C_0 + C_1 * \text{size}(R) * \text{card}(R')$
- Cost of computing Q at site – 1:
 $TC_4 = ?$



Semi-join Programs (contd.)

Cost of Semi-join program

- Cost of sending S' to site – 2:
 $TC_1 = C_0 + C_1 * \text{size}(A) * \text{val}(A[S])$
- Cost of computing R' at site – 2:
 $TC_2 = 0$
- Cost of sending R' to site – 1:
 $TC_3 = C_0 + C_1 * \text{size}(R) * \text{card}(R')$
- Cost of computing Q at site – 1:
 $TC_4 = 0$



Semi-join Programs (contd.)

Cost of Semi-join program

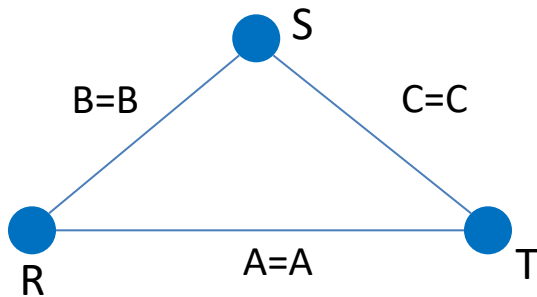
- Total cost $TC_{SJ} = TC_1 + TC_2 + TC_3 + TC_4$
- If $TC_{SJ} < TC_{JN}$ then semi-join program is profitable.
 - Here TC_{JN} is the cost of performing join without semi-join program.

Other Applications of Semi-join Programs

- Semi-join programs can be used as fragment reducers (operations that can reduce cardinality of a relation).
 - Similarly to unary operations.
- Full reducer:
 - Chain of semi-joins.

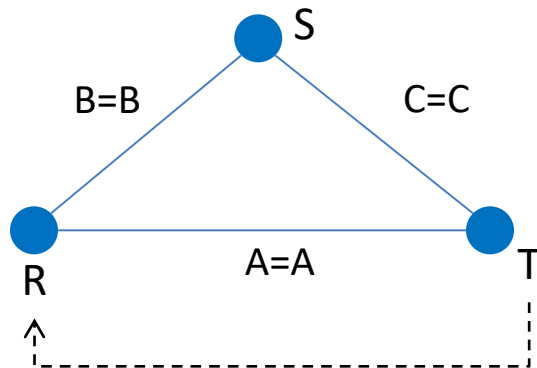
Full Reducer

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4



Full Reducer (contd.)

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4

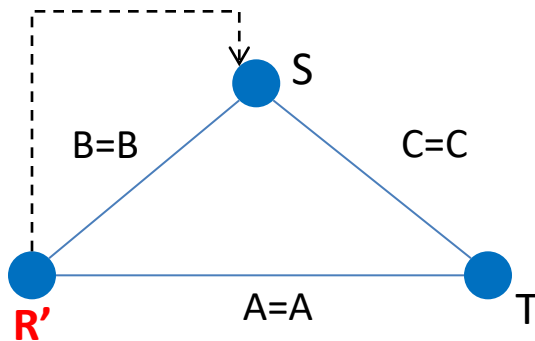


$$R' = R S J_{A=A} T$$

A	B
2	b
3	c

Full Reducer (contd.)

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4



$$\mathbf{R}' = \mathbf{R} \mathbf{S} \mathbf{J}_{A=A} \mathbf{T}$$

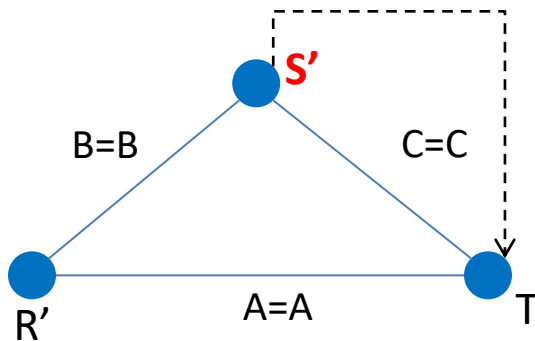
A	B
2	b
3	c

$$\mathbf{S}' = \mathbf{S} \mathbf{S} \mathbf{J}_{B=B} \mathbf{R}'$$

B	C
b	y
c	z

Full Reducer (contd.)

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4



$$R' = R \text{ SJ}_{A=A} T$$

A	B
2	b
3	c

$$S' = S \text{ SJ}_{B=B} R'$$

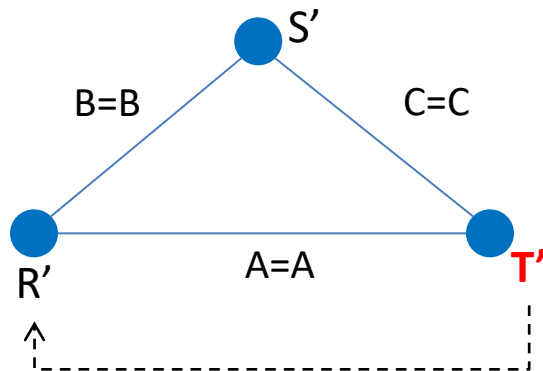
B	C
b	y
c	z

$$T' = T \text{ SJ}_{C=C} S'$$

C	A
y	3
z	4

Full Reducer (contd.)

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4



$$R' = R \text{ SJ}_{A=A} T$$

A	B
2	b
3	c

$$S' = S \text{ SJ}_{B=B} R'$$

B	C
b	y
c	z

$$T' = T \text{ SJ}_{C=C} S'$$

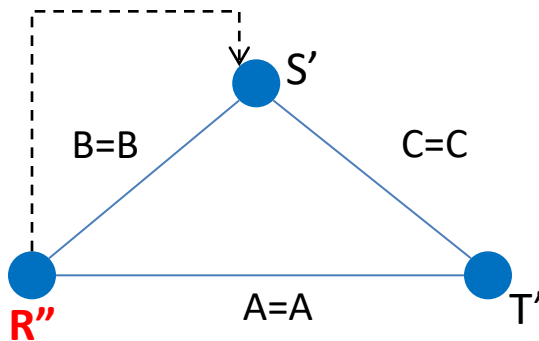
C	A
y	3
z	4

$$R'' = R' \text{ SJ}_{A=A} T'$$

A	B
3	c

Full Reducer (contd.)

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4



$$R' = R \text{ SJ}_{A=A} T$$

A	B
2	b
3	c

$$S' = S \text{ SJ}_{B=B} R'$$

B	C
b	y
c	z

$$T' = T \text{ SJ}_{C=C} S'$$

C	A
y	3
z	4

$$R'' = R' \text{ SJ}_{A=A} T'$$

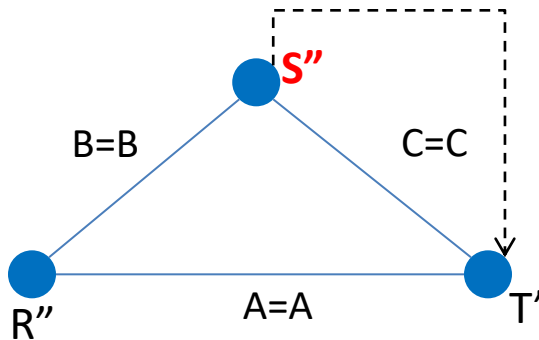
A	B
3	c

$$S'' = S' \text{ SJ}_{B=B} R''$$

B	C
c	z

Full Reducer (contd.)

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4



$$R' = R \text{ SJ}_{A=A} T$$

A	B
2	b
3	c

$$S' = S \text{ SJ}_{B=B} R'$$

B	C
b	y
c	z

$$T' = T \text{ SJ}_{C=C} S'$$

C	A
y	3
z	4

$$R'' = R' \text{ SJ}_{A=A} T'$$

A	B
3	c

$$S'' = S' \text{ SJ}_{B=B} R''$$

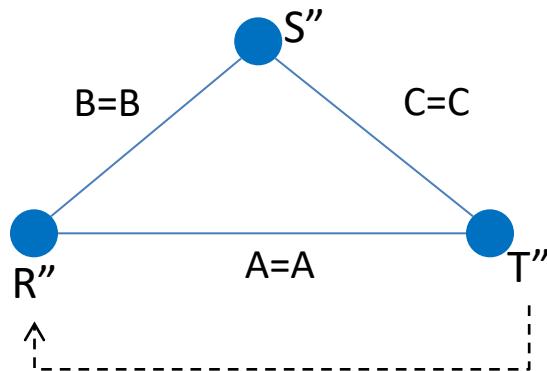
B	C
c	z

$$T'' = T' \text{ SJ}_{C=C} S''$$

C	A
z	4

Full Reducer (contd.)

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4



$$R' = R \text{ SJ}_{A=A} T$$

A	B
2	b
3	c

$$S' = S \text{ SJ}_{B=B} R'$$

B	C
b	y
c	z

$$T' = T \text{ SJ}_{C=C} S'$$

C	A
y	3
z	4

$$R'' = R' \text{ SJ}_{A=A} T'$$

A	B
3	c

$$S'' = S' \text{ SJ}_{B=B} R''$$

B	C
c	z

$$T'' = T' \text{ SJ}_{C=C} S''$$

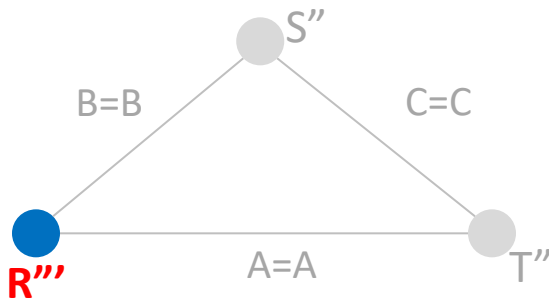
C	A
z	4

$$R''' = R'' \text{ SJ}_{A=A} T''$$

Φ

Full Reducer (contd.)

R		S		T	
A	B	B	C	C	A
1	a	a	x	x	2
2	b	b	y	y	3
3	c	c	z	z	4



$$R' = R \text{ SJ}_{A=A} T$$

A	B
2	b
3	c

$$S' = S \text{ SJ}_{B=B} R'$$

B	C
b	y
c	z

$$T' = T \text{ SJ}_{C=C} S'$$

C	A
y	3
z	4

$$R'' = R' \text{ SJ}_{A=A} T'$$

A	B
3	c

$$S'' = S' \text{ SJ}_{B=B} R''$$

B	C
c	z

$$T'' = T' \text{ SJ}_{C=C} S''$$

C	A
z	4

$$R''' = R'' \text{ SJ}_{A=A} T''$$

Φ

Additional Reading

- Length of full reducer.
- Tree queries.
- “Best” reducer.

Practice Problems/ Questions

1. With an example, prove that the semi-join is not symmetric.
[hint: page. 142]
2. Textbook exercise: 6.1, 6.4